

LAB ASSIGNMENT-03

- **Problem Statement:** Write a Java Program to implement method overloading & method overriding in Java.
- i) Write a class shapes. Find the area of basic shapes using polymorphism.
- ii) Write a Java program to create a parent class Hillstations & 3 subclasses by hill station names. Subclasses extend the superclass & override methods famous food() & famous for(). Call them by parent class. It should refer to base class object & baseclass method overrides superclass method.

- **Objective :** i) To study method overloading.
- ii) To study method overriding in Java.

- **Theory :**
- i) **Method overloading & its advantages :**
 - It is a feature in Java where multiple methods have same name but different parameters (type, number, or order) within same class.
 - Advantages :-**
 - Improves readability ~~code~~ of code.
 - Increases code reusability.
 - Reduces need for multiple method names.
- ii) **Method overriding & its rule :**
 - Occurs when a subclass provides a specific implementation of a method already defined in its superclass.
 - Rules :-**
 - Method must have same named parameters.
 - Must be in inheritance relationship.

- Return type must be same or covariant.
- Can't override final, static, or private methods.

• Algorithm :-

- Start
- Create a class with multiple methods having same name but different parameters.
- Create a parent class with a method.
- Create a child class that extends parent class.
- Override the parent class method in child class.
- Create objects & call methods.
- Display outputs.
- End.

• Conclusion :

Thus, we've successfully implemented method overloading & overriding.

• FAQs :-

Q.1 Can we overload Java main() method?

Ans Yes, the main() method can be overloaded in Java by changing its parameters but JVM will only call :-

```
public static void main(String[] args)
```

Q.2 Can we declare overloaded methods as final?

Ans Yes, overloaded methods can be declared as final.
As, final prevents overriding not overloading.

Q.3 Can an overloaded method be overridden?

5: Yes, an overloaded method can also be overridden in a subclass.

→ First, methods are overloaded in a class

→ Then, a subclass can override one of those methods.

What is method overriding?

Method overriding is a concept where a child class redefines a method of its parent class to provide a specific implementation.

Characteristics :-

i) It requires ~~inheritance~~ inheritance.

ii) It happens at runtime.

iii) Uses dynamic method dispatch.

5: What happens if we change the arguments of overriding method?

Overriding requires exact same method signatures.

→ If the method parameters are changed, it is treated as a new method & becomes method overloading.

6: Can we change the return type of overriding method from number type to integer type?

Yes, this is allowed & is called a covariant return type.

→ The return type in the sub-class can be a subclass of the parent's return type.

Eg :-

Number getValue() { return 10; }

Integer getValue() { return 20; }